



KF-7434

Integrated M. C. A. (Sem. III)
Examination, December-2024

IMCA-304T

STATISTICS AND PROBABILITY

Time allowed : Three Hours

Max. Marks : 70

Min. Marks : 28

*This question paper contains two sections
as under:*

SECTION - A

Max. Marks-20

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×2=20)

SECTION - B

Max. Marks-50

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×5=50)

Note : Use of Non-programmable scientific calculator is allowed in this paper.

SECTION - A

- 1 (i) Define random variable and sample space.
- (ii) Write down the two properties of probability distribution function.
- (iii) Write down the pdf for Poisson distribution and Geometric distribution.
- (iv) Write down the Mean and Variance of Binomial distribution.
- (v) Write down the pdf of Normal distribution. Also write its Mean and variance.
- (vi) Write down the Mean and Variance of Uniform distribution.
- (vii) Define Correlation and Regression.
- (viii) Write down the formula for calculating Spearman's Rank Correlation coefficient? What is the range of values of correlation coefficient?
- (ix) Define method of least squares for fitting a curve.
- (x) Write down the normal equations for fitting a second degree parabola.

SECTION - B

UNIT - I

- 2 A doctor is to visit a patient. From past experience it is known that the probabilities that he will come by train, bus, scooter or any other means of transport are $3/10$, $1/5$, $1/10$ and $2/5$ respectively. The probabilities that he will be late are $1/4$, $1/3$ and $1/12$ if he comes by train, bus and scooter respectively. If he comes by other means of transport, he will not be late. When he arrives, he is late. What is the probability that he came by train?

OR

In an experiment of rolling two fair die, let X be the random variable denoting the sum of numbers that appear on the two die. Find $E(X)$, $E(X^2)$ and $\text{Var}(X)$.

UNIT - II

- 3 (i) Derive the Mean and Variance of Binomial distribution.
(ii) Derive the pdf of Poisson distribution as limiting case of Binomial distribution.

OR

Derive the formula for recurrence relations for Moments about Origin and Moments about Mean for Poisson distribution.

UNIT - III

- 4 (i) Derive Mean and Variance of Exponential distribution.
(ii) If X has uniform distribution in $[0, 1]$ with pdf $f(x) = 1$, $0 \leq x \leq 1$. Find the pdf for $2 - \log(x)$.

OR

If X is a Normal Variate with Mean = 30 and Standard Deviation = 5, find the probability that

(i) $26 \leq X \leq 40$

(ii) $X \geq 45$

(iii) $|X - 30| \leq 5$

{Given z -table values: $z(0.8) = 0.2881$, $z(1) = 0.3413$, $z(2) = 0.4772$, $z(3) = 0.49865$ }

UNIT - IV

- 5 Calculate the Karl Pearson's correlation coefficient between X and Y for the given data:

X : 1 2 3 4 5 6 7 8 9

Y : 9 8 10 12 11 13 14 16 15

OR

In a partially destroyed laboratory record of an analysis of correlation data, the following results are only legible:

Variance of X is 9,

Regression equations are $8x - 10y + 66 = 0$ and $40x - 18y = 214$

Find:

(i) Mean values of X and Y

(ii) Variance of Y

(iii) Coefficient of correlation between X and Y

UNIT - V

- 6 Fit a straight line to the following data:

X : 1 2 3 4 5 6 7 8 9

Y : 2 6 7 8 10 11 11 10 9

OR

Fit a parabola of second degree to the following data:

X : 0 1 2 3 4

Y : 1 5 10 22 38